Huaiyuan Yao

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Google Scholar: 209 citations, h-index 8 (Nov. 2026)

Education

Arizona State University, U.S. Fall 2025 – present

Ph.D. in Computer Science, SCAI. Advisor: Prof. Hua Wei (DaRL Lab). Expected graduation: Spring 2029. Research interest: AI agents that keep learning from their own experience after real-world deployment.

Xi'an Jiaotong University, China 2021 – 2025

B.S. in Computer Science. Advisor: Prof. Yuanqi Su.

Research Experience

Honda Research Institute USA, Mountain View, CA May – Aug 2026

Research Intern (mentor: Manish Saroya). LLM-steered deterministic driving planners (HarnessDriver).

DaRL Lab, Arizona State University 2024 – present

Graduate Research Associate (2025 – present); *Research Intern*, remote (2024 – 2025). CoMAL, Instructional Agents, self-evolving multi-agent LLMs (LangMARL, MASkills), and test-time training.

Institute for AI Industry Research (AIR), Tsinghua University, China 2024 – 2025

Research Assistant (advisors: Prof. Peng Li, Prof. Yilun Chen). LiloDriver: lifelong motion planning.

Publications (7 first-authored)

Peer-reviewed conference papers

[1] **Huaiyuan Yao**, Longchao Da, Xiaoou Liu, Charles Fleming, Tianlong Chen, Hua Wei. *LangMARL: Natural Language Multi-Agent Reinforcement Learning via Semantic Credit Assignment.* EMNLP 2026 (Main).

[2] **Huaiyuan Yao**, Xiaoou Liu, Charles Fleming, Tianlong Chen, Hua Wei. *MASkills: Continual Skills Optimization for Multi-Agent LLM Systems.* Findings of EMNLP 2026.

[3] **Huaiyuan Yao**, Wanpeng Xu, Justin Turnau, Nadia Kellam, Hua Wei. *Instructional Agents: Reducing Teaching Faculty Workload through Multi-Agent Instructional Design.* EACL 2026. GitHub: 500 stars.

[4] **Huaiyuan Yao**, Zhenxiao Cao, Zhongman Wang, Xiewei Ni, Jinyan Dong, Hongyi Cai, Yuqiang Han, Xiao Luo. *How Effective is In-Context Learning with Large Language Models for Rare Cell Identification in Single-Cell Expression Data?* IEEE ICDM 2025.

[5] **Huaiyuan Yao**, Longchao Da, Vishnu Nandam, Justin Turnau, Zhiwei Liu, Linsey Pang, Hua Wei. *CoMAL: Collaborative Multi-Agent Large Language Models for Mixed-Autonomy Traffic.* SDM 2025, **Best Poster**.

[6] Tiejin Chen, Maunil Rohitbhai Vyas, **Huaiyuan Yao**, Hua Wei. *Steering Vectors as a Training Signal in LLM Post-Training.* NeurIPS 2026.

[7] Tiejin Chen, **Huaiyuan Yao**, Jia Chen, Evangelos E. Papalexakis, Hua Wei. *Every Response Counts: Quantifying Uncertainty of LLM-based Multi-Agent Systems through Tensor Decomposition.* ACL 2026.

[8] Yihan Hong, **Huaiyuan Yao**, Bolin Shen, Wanpeng Xu, Hua Wei, Yushun Dong. *From Rubrics to Reliable Scores: Evidence-Grounded Text Evaluation with LLM Judges.* EMNLP 2026 (Main).

[9] Ziqi Wang, **Huaiyuan Yao**, Hua Wei, Hang Zhou, Xiao Luo. *AURORA: Attention-Intended Dual Verification with Adaptive Rollback for LLM Agents.* IROS 2026.

[10] Jacob Igo, **Huaiyuan Yao**, Wanpeng Xu, Nadia Kellam, Hua Wei. *From Course Concept to Lecture Video: An AI-Powered System for Automated MOOC Development.* LERN 2026.

Preprints and manuscripts

[11] **Huaiyuan Yao**, David Isele, Sangjae Bae, Hua Wei, Manish Saroya. *HarnessDriver: Steering Deterministic Driving Planners toward Adaptive Closed-Loop Behavior.*

[12] **Huaiyuan Yao**, Pengfei Li, Bu Jin, Yupeng Zheng, An Liu, Lisen Mu, Qing Su, Qian Zhang, Yilun Chen, Peng Li. *LiloDriver: A Lifelong Learning Framework for Closed-Loop Motion Planning in Long-Tail Autonomous Driving Scenarios.* arXiv:2505.17209, 2025.

[13] Longchao Da, Tiejin Chen, Zhuoheng Li, Shreyas Bachiraju, **Huaiyuan Yao**, Li Li, Yushun Dong, Xiyang Hu, Zhengzhong Tu, Dongjie Wang, Yue Zhao, Ben Zhou, Ram Pendyala, Benjamin Stabler, Yezhou Yang, Xuesong Zhou, Hua Wei. *Generative AI in Transportation Planning: A Survey.* arXiv:2503.07158, 2025.

[14] Lina Zhao, Zihao Bian, Qingyue Chen, Yafang Li, Zhiyi Luo, Jiaxing Bai, Guangbo Li, Min He, Kezhi Li, **Huaiyuan Yao**†, Zongjiu Zhang. *Autonomous Multi-Modal LLM Agents for Treatment Planning in Focused Ultrasound Ablation Surgery.* arXiv:2505.21418, 2025. († corresponding author)

[15] Ferdous Al Rafi, Susrik Mukherjee, Latika Liladhar Dekate, Jennifer Yawa Lavoe, **Huaiyuan Yao**, Shlok Mohanty, Longchao Da, Xuesong Zhou, Hua Wei. *Sim2Signal: Sim-to-Real Benchmarks for Traffic Signal Control.* arXiv:2609.01676, 2026.

[16] Wenwen Li, Pingping Wang, Yuanyuan Tian, Sizhe Wang, **Huaiyuan Yao**, Longchao Da, Al Mehdi Saadat Chowdhury, Mahdi Ahmadi, Yi-Wen Mo, Ferdous Al Rafi, Hua Wei, YooJung Choi, Paulo Shakarian. *LifelineQA: An Evidence-Grounded Benchmark for Situational-Awareness Question Answering in Emergency Operations.*

[17] Qingyue Chen, **Huaiyuan Yao**, Wanpeng Xu, Varun Shourie, Qianwen Wang, Hua Wei. *MAVE: A Multi-Agent Framework for Grounded Pedagogical Feedback from Online Class Videos.* arXiv:2610.24498, 2026.

★ Grants, Fellowships, and Honors

NSF ACCESS allocation, Principal Investigator: test-time training for self-improving LLM agents — 2026

Google TPU Research Cloud (TRC): TPU compute support — 2026

Best Poster Award, SIAM International Conference on Data Mining (SDM), for CoMAL — 2025

University Graduate Fellowship, Arizona State University — Spring and Summer 2026

Global Security Initiative (GSI) Fellowship, Arizona State University — Fall 2025

Selected participant, AI-SCORE 2026 workshop on OR and CS, UIUC — 2026

Finalist Prize, Mathematical Contest in Modeling (MCM), COMAP — 2023

🏛 Service, Teaching, and Mentoring

- **Reviewer:** AAAI 2027 Student Abstract and Poster Program (PC), ECML-PKDD 2026, RLC 2026, IROS 2026, ACML 2026, WSDM 2026, EWRL 2026, CIKM 2026, RV 2026, ICMLA 2026, PAKDD 2026.
- **Community:** Volunteer, China3DV 2025; Founder, Neural Modeling and Programming Workshop (2024), a student-run computational neuroscience workshop with hundreds of participants.
- **Teaching Assistant, ASU:** CSE 598 Agentic AI (Fall 2026); CSE 475 Foundations of Machine Learning (Fall 2025), with Prof. Robert K. Atkinson; CSE 220 Programming for Computer Engineering (Spring 2026), with Prof. Adwith Malpe.
- **Mentoring:** undergraduate and master's students Jacob Igo, Siyuan Xing, Aarshdeep Nandra, Yuxuan Huang, Rayan Kazi, Qingyue Chen (ASU), Ziqi Wang (Rutgers), Iris Hong (WashU).
- **Open source:** Instructional Agents (500 GitHub stars), CoMAL, LiloDriver, LangMARL, MASkills.